

AIM: Implementing Random Forest to check the weather data.

1.

CODE:

```
import random

def generate_weather_data(years,num_locations):
    data=[]
    for year in range(years):
        for location in range(num_locations):
            for day in range(365):
                for hour in range(24):
                    temperature=round(random.uniform(-20.0,40.0),1)
                    data.append((year+2000,location,day+1,hour,temperature))
    return data

def filter_data(data,threshold):
    filtered_data = [record for record in data if record[4] > threshold]
    return filtered_data

def calculate_statistics(data):
    if not data:
        return 0,0,0
    average_temp=sum(record[4]for record in data)/len(data)
    max_temp = max(record[4] for record in data)
    min_temp = min(record[4] for record in data)
    return average_temp, max_temp, min_temp

def filter_and_write_to_file(data,threshold,output_file):
    with open(output_file,'w') as f:
        for record in data:
            if record[4] > threshold:
                f.write(','.join(map(str, record)) + '\n')

sample_data=generate_weather_data(5,10)

yearly_data = {}

for record in sample_data:
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year = record[0]

if year not in yearly_data:
    yearly_data[year] = []

yearly_data[year].append(record)

for year, data in yearly_data.items():
    avg_temp, max_temp, min_temp = calculate_statistics(data)

    print(f"Year: {year}, Average Temperature: {avg_temp:.2f}, Max Temperature: {max_temp}, Min
Temperature: {min_temp}")

filtered_data = filter_data(sample_data, 30.0)

filter_and_write_to_file(filtered_data, 30.0, 'filtered_data.txt')
```

OUTPUT:

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===== RESTART: C:\Coding Files\Python313\Python.py =====
Year: 2000, Average Temperature: 9.96, Max Temperature: 40.0, Min Temperature: -20.0
Year: 2001, Average Temperature: 9.92, Max Temperature: 40.0, Min Temperature: -20.0
Year: 2002, Average Temperature: 9.96, Max Temperature: 40.0, Min Temperature: -20.0
Year: 2003, Average Temperature: 10.06, Max Temperature: 40.0, Min Temperature: -20.0
Year: 2004, Average Temperature: 9.94, Max Temperature: 40.0, Min Temperature: -20.0
|
```